Sound-Activated LED Alarm

By Matthew Tran and Timothy Chegal

Lab Section 202

Man-Hours

Researching the design of the circuit took about 3-4 hours since there were almost no designs online that consisted of two operational amplifiers and two capacitors. The designs from different references/sources were combined to draft the design on the circuit. Building the circuit took about 2.5 hours on Tuesday, 4 hours on Wednesday, and more at home for slight modifications and clean-up. The total man hours is about 11-12 hours.

Cost Breakdown

Component	Price
Electret Microphone (8-pack)	$8.93
LF356N Operational Amplifier (2x Quantity)	ECE 285 Lab Kit
1uF Capacitor	ECE 285 Lab Kit
4.7nF Capacitor	ECE 285 Lab Kit
1kΩ Potentiometer	ECE 101 Lab Kit
Red LED	ECE 285 Lab Kit
Breadboard, Resistors, Wires	ECE 101/285 Lab Kit

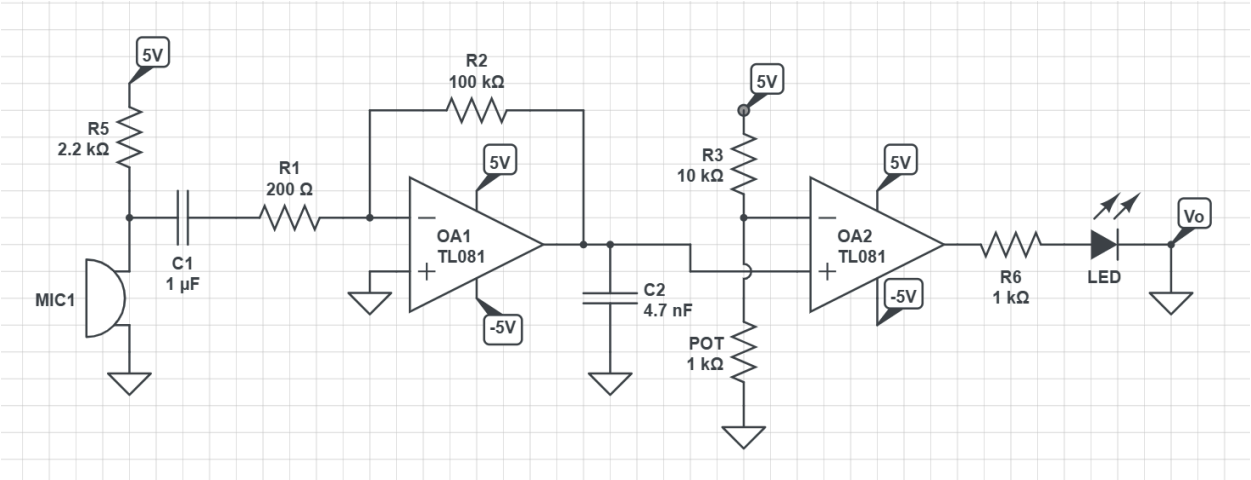
*The ECE 285 Lab Kit was $10, and the ECE 101 Lab Kit was about $50.

Circuit Description

The main components of the sound-activated alarm consist of an electret microphone, two op-amps, two capacitors, 1kΩ potentiometer, and LED. The first op-amp is used as an inverting amplifier that amplifies the input signal from the electret microphone which is usually around tens to hundreds of millivolts. The second op-amp is a comparator which plays a big role in the voltage activation threshold of the LED. One capacitor (1uF) was placed before the inverting amplifier input as a high-pass filter to block biased DC signals coming from the power supply and pass AC signals through to the inverting amplifier. The other capacitor (4.7nF) was placed after the output of the inverting amplifier, and it was placed to ground in parallel to the comparator so that any excess noise from the amplified signal would be filtered before it entered

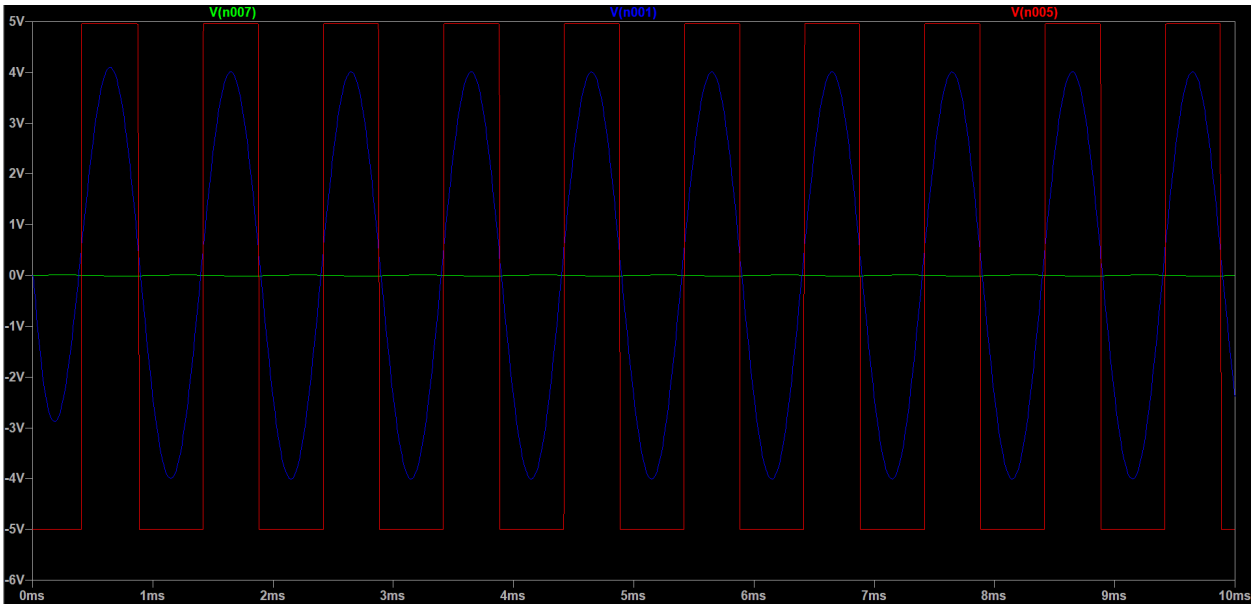
the comparator. The amplified signal went into the positive input of the comparator while a constant input voltage was applied to the negative input. The voltage to the negative input can be adjusted by increasing or decreasing the potentiometer resistance. Since the amplified signal is being compared to a constant voltage, the voltage activation threshold of the circuit is changed by adjusting the potentiometer. If the voltage of the amplified input signal is greater than the constant activation threshold, then the comparator will output a positive voltage to turn the LED on, and it will output a negative voltage to keep the LED off if the threshold is not met.

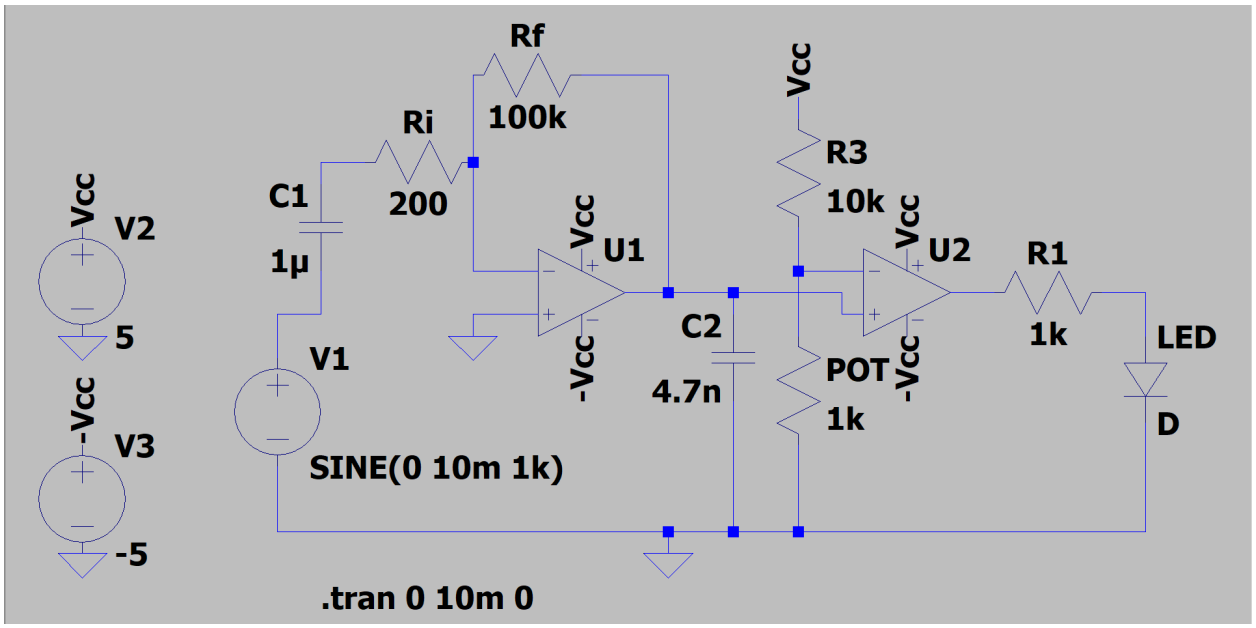
Circuit Schematic



Circuit Simulation + Hand Analysis

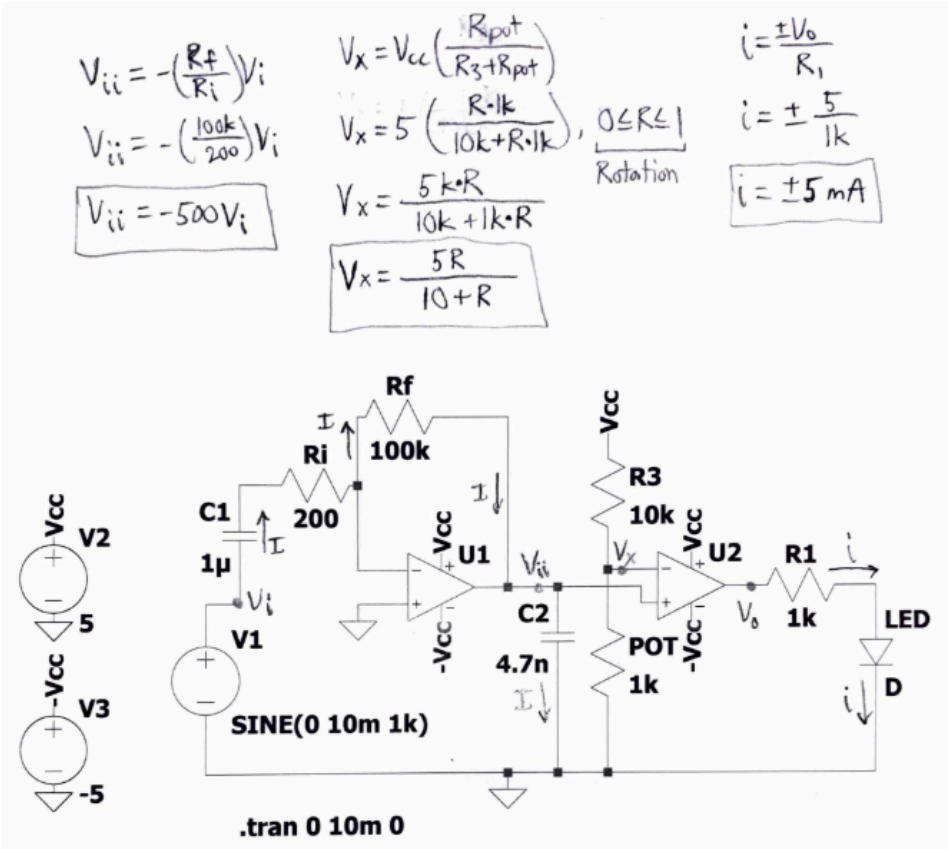
LTspice Simulation





From the simulation graph above, the green signal represents the input voltage signal from the electret microphone. The blue signal is the amplified input signal that came from the inverting amplifier. The red signal is the output voltage signal that came from the comparator.

Hand Analysis



Power Consumption

The total power consumption of the sound-activated alarm includes the voltage and current coming into the power rails of the inverting amplifier and comparator. Using two channels from a power supply, a positive $5V$ was provided to the positive rails of the op-amps, and a negative $5V$ was provided to the negative rails of the op-amps. That means a total magnitude of $10V$ was provided to the entire circuit. When the power supply was turned on (which activated the circuit), the current through each channel automatically adjusted to $0.02A$, so a total of $0.04A$ was provided to the circuit. Power is defined by $P = VI$, so multiplying the total voltage and total current ($10V * 0.04A$) results in a power consumption of 0.4 Watts.

Member Contribution

Matthew researched the possible designs for the sound-activated alarm and made sure the design included two op-amps and capacitors. He researched how capacitors could be effective filters in a sound-activated circuit that uses an AC signal and how they could be incorporated with the op-amps. He created and assembled the initial circuit design and made necessary changes to the design along the way to expand the usability of the circuit.

Timothy made sure to assist with the building by troubleshooting all the breadboard concepts and components to make sure the project was functioning properly, and he simulated the circuit on LTspice. Timothy also was in charge of acquiring the needed components that were not included with the lab kit. Lastly, Timothy was in charge of the research for the back up project: “Treble and bass audio filter” and made sure it also included two op-amps alongside with two capacitors in case the main project was not working at all.

Challenges

There were three main challenges: First, our main issue was figuring out the appropriate configurations for our project. We fixed this by switching our non-inverting op-amp into an inverting one to correctly amplify the sound.

Secondly, our circuit’s rails weren’t properly wired up causing incorrect flow. We fixed this by using both positive and negative 5V for the rails and hooked them up to a common ground to ensure correct current flow throughout the circuit.

Finally, the comparator lacked sensitivity and was unable to produce a high output because the amplified signal never exceeded the reference voltage set by the voltage divider. To address this, we reduced the voltage divider resistance by replacing the 10 kΩ resistor with a 1 kΩ potentiometer. This not only allowed the comparator to switch properly but also introduced adjustability, enabling us to control the sensitivity of the circuit by tuning the potentiometer.

Potential Improvements

There are several ways the circuit could be improved. First, a buffer stage could be added after the voltage divider. This would isolate the high-impedance source from the comparator, providing a more stable and cleaner signal. Additionally, the buffer would prevent loading effects and minimize voltage drops, improving overall signal integrity.

Second, although the potentiometer provided useful adjustability, it was somewhat inconvenient to fine-tune. A more practical solution would be to implement a switchable resistor network consisting of multiple resistors with different values connected to ground. By selecting different resistors using a switch, the user could quickly adjust the sensitivity of the circuit. For example, selecting a lower resistance would increase sensitivity, while selecting a higher resistance would decrease it.

Applications

Although this circuit is a simplified design, it demonstrates the fundamental principles used in many real-world voice-activated systems. Similar concepts can be found in devices such as Amazon Alexa, Google Home, Apple Siri, and Ring, as well as in various audio and music processing equipment. These systems rely on detecting and amplifying sound signals, followed by threshold-based decision-making, much like the functionality demonstrated in this project

References

Circuit Design

- https://electronics.stackexchange.com/questions/339333/sound-detector-using-opamp
- https://www.youtube.com/watch?v=mkGr5_LMkhM

Datasheets

- LN356N Operational Amplifier: https://www.ti.com/lit/ds/symlink/lf356.pdf?ts=1777584070938&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FLF356

Parts Purchased

- Electret Microphone: https://www.amazon.com/gp/product/B01F5592EO/ref=ox_sc_act_title_1?smid=ATVPDKIKX0DER&psc=1